

The Virtual Science Fair

HCI Knowledge about Information Design

Gestalt Principle		User interface examples
Proximity	elements near each other tend to be seen as a group	words on a menu bar, columns in a tabular display, text in a paragraph
Similarity	elements that share visual characteristics (shape, color, etc.) tend to be seen as a group	toolbar icons, data visualization
Closure	there is a tendency to organize elements into complete, closed figures	overlapping windows, menus, dialog boxes, and other user interface controls
Area	there is a tendency to group elements in a way that creates the smallest possible figure	icons on a workstation screen, pop-up menu on top of a document display
Symmetry	there is a tendency to see symmetric elements as part of the same figure	window manipulation controls (e.g., scroll bar, selection handles)
Good continuation	there is a tendency to group elements into continuous contours or repeating patterns	a page of paragraphs, a grid of spreadsheet cells, a left-justified list of selections or parameters

Information Metaphors

SF Information	Real World Metaphor	Implications for VSF Information Design
An exhibit looks like a...	Lab journal	Loosely organized pages with handwriting, sketches
	Documentary	Movie or animated sequence of screens and audio
A teacher-coach looks like a...	Peer (colleague)	Friendly face of same age, character as student

	Director	Professional-looking image with specific tools, agenda
The fair looks like a...	Study room	Empty work area with a place to write, materials to be read
	Public lecture	Uninterrupted single presented in focal view
	Cocktail party	Groups of people, auditory cues to conversation, activity
A judging form looks like a...	Balance sheet	Grid with cells displaying labels, values, equations
	Discussion	Questions, answers, comments in sequential order
A fair summary looks like a...	Report card	Checkmarks, letters, or numbers marking assessment scales
	Guided tour	A sequence of exhibits with annotations or comments
	Thank you note	Form letter with personal salutation, thanks, signature

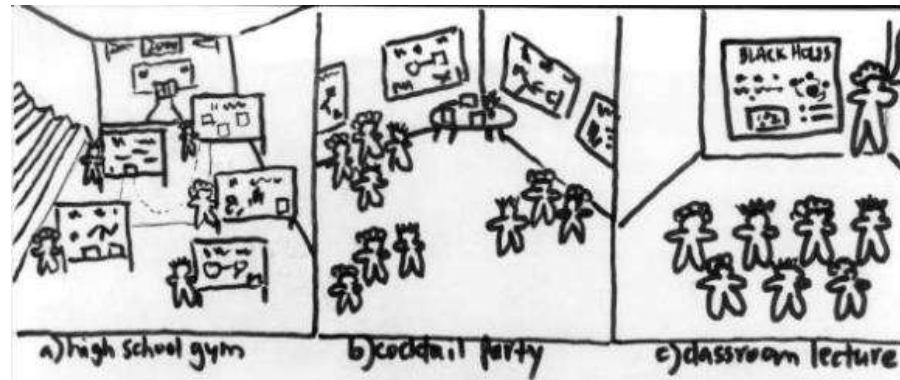
Information Technology options

Artifact

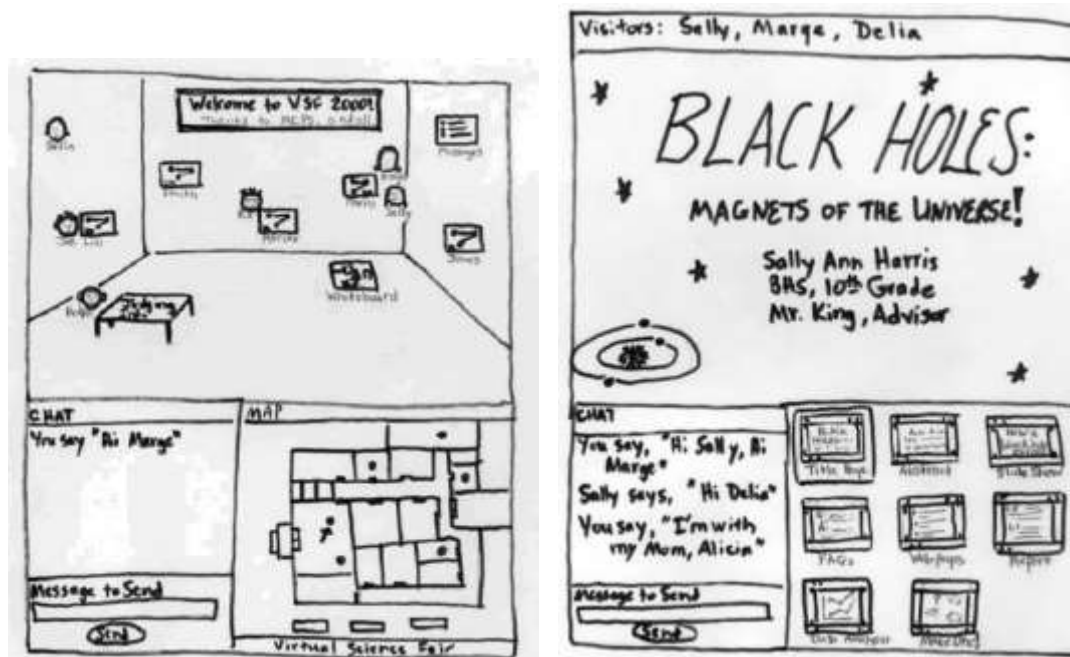
VSF Information	MOOsburg Technology	Implications for VSF Information Design
An exhibit looks like a...	Multimedia notebook	A video and audio presentation organized into pages

	Electronic Whiteboard	Rectangular white space with colored lines and text
	Web pages	Netscape-like browser with underlined hot links
A teacher-coach looks like a...	Email	A list of messages organized by date or sender
	Threaded discussion	A list of comments indented to show replies/responses
	Chat	Sequential log of messages prefaced with sender name
The fair looks like a...	Room panorama	Panorama of walls, floor, maybe doors, and windows
	Interactive map	Geographic coordinates showing landmarks, paths
A judging form looks like a...	Voting booth	Privacy curtain with levers or buttons labeled with choices
	Threaded discussion	Exhibits at top level, evaluative comments/replies indented
A fair summary looks like a...	Charting package	Labeled X and Y axis with lines, histograms, etc.
	Multimedia notebook	A video and audio presentation organized into pages
	Interactive map	Path traced out on a map, showing notable places to visit

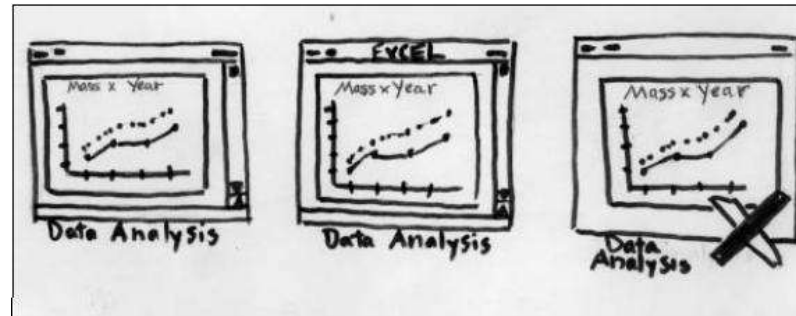
Design sketch



Key Screens



Detailed design sketch



Information Scenarios

[1] Sally plans her exhibit on black holes.

- Sally's new exhibit opens as a [separate window](#) next to the view of the VSF.
- She sees that it already has a [pre-defined structure](#), with placeholders for what seem to be exhibit components. There is a main view with a number of [smaller views](#) surrounding it.
- The main view shows "[Title page image](#)" in a fancy graphical font. She infers she should import an image file to use as her title page. She also infers that this view is displayed when the small "title" window is selected.
- When she [explores the other miniatures](#) she sees other prompts, for example "Powerpoint slide show" and "Project report". She doesn't see anything about simulations, so worries whether she can include hers.
- She sees a [list of visitors](#), now only herself, and some exhibit tools—a project calendar and discussion board. Will she have to add her simulation as a new exhibit tool?
- She adds the image file from her PC that she created as a title page. As she does this, she notices menu options for renaming, deleting, and adding exhibit components. Relieved, she realizes that the standard layout is just a starting point. She now is confident that she can add her simulation as a component.

[2] Mr. King coaches Sally on her project.

- Mr. King checks Sally's project calendar, and he sees from the grayed out sections that she is busy all day. So he adds an appointment to work with her tonight.
- He sees that Sally is already working when he arrives. He joins her and sees that she has added a title page, and is reviewing her

project slideshow. He opens the slide show to watch along, using the chat tool to make occasional comments.

- When they finish, she tells him she will now bring in some other files from her PC. He watches while she adds a project notebook and a set of data analyses.
- In the miniature, the data analysis looks like an Excel spreadsheet, and he wonders if it is read-only or if it will invoke Excel. He tries a double-click, and it opens the data in Excel in a separate window. He worries that visitors might tamper with Sally's data, but she explains that he can make changes but not save them.
- Mr. King is concerned that Sally is bringing in too many components. He reminds her that even though this is an unlimited space, visitors and judges do not have unlimited time, and will be biased by first impressions. Sally agrees and they start removing components that do not reveal much at first glance.
- When Sally shows her idea about the interactive simulation, he worries that visitor-contributed results will make her display too messy. He explores the exhibit construction menu and finds a way to create [nested components](#). He suggests using layers to hold visitor-created content, and other less central materials.

[3] Alicia and Delia go to the science fair.

- The email includes a string that Delia recognizes as a URL in MOOsburg.
- At the VSF they recognize standard [MOOsburg layout-panorama view](#) of the fair, brief list of objects to work with, chat tool, interactive map.
- Alicia recognizes map as a high school floor plan. She shows Delia where she worked in the office as a peer counselor. They see a green dot in the gym, blue dots in other rooms. Alicia infers they are "in" the gym; she plans to check out the rest later.
- The main view is crowded. At the back is a large Welcome sign, with thanks to organizers, other announcements.
- Exhibits are arrayed around the room, each with a student name attached. Some are covered with a [black and yellow banner](#); Delia suggests that these must be "under construction".
- People are in the room. Some are small photos, others smiley faces or simple line drawings. Because they logged in under Delia's school ID, they appear as her school photo.
- They are attracted to an exhibit with lots of people around, but then Alicia notices friend Marge, so they join her instead at Sally's exhibit.
- The exhibit appears in a [separate window](#). Like in the welcome area, there is a main view. It displays what looks to them like a title page.
- Near the main view what appear to be [several miniaturized windows](#). There is also a [list of visitors](#) and a text chat window.
- Sally and Marge's conversation is being displayed in the chat log; they pause briefly to say hi.
- Alicia is impressed with how much Delia knows about how to use PowerPoint and Excel. Delia explains she has been using these in science. They watch the slide show for an overview, as Sally's voice-over explains key points.
- Delia sees a discussion board with an indented list of questions and answers. Alicia enjoys seeing her study the questions and pose

one of her own.

- The simulation is unfamiliar to Alicia, so she asks Sally. She demos it and invites them to change it, showing them the "Visitor Experiments" folder to use for their results.

[4] Ralph judges the high school physics projects at the science fair.

- When he first visits, he sees a panorama with many exhibits around the room. He is surprised at how many still show "under construction" and worries whether they can all be done in time for judging on Thursday.
- On Thursday, he logs in using the special judging ID provided by Rachel. He appreciates how this gives him a [custom view of the VSE](#), showing just the five projects he has been assigned, along with the judging forms.
- The judging form looks a lot like the paper forms he has used for years. He does notice that the numerical ratings look like input fields, and hopes this means the [final calculations](#) will be done for him.
- He wants a quick overview, and is delighted to find that he can open all five exhibits at once, even though the display becomes complex when he also opens all of the forms to take notes.
- Later, he comes back to examine the projects in more detail, and as in the past finds the form inadequate.
- He wonders if he can weight the categories in the two forms. He sees that the final score is a formula and he is [allowed to edit it](#), weighting Jeff's and Sally's ratings differently. But his change cannot be saved until he adds a note explaining his rationale for the modifications.
- After submitting all five forms, a new object appears in the VSF panorama, labeled "Judging results". He opens it and sees a set of histograms charting results in each age group. Sally and Jeff are tied!

[5] Rachel organizes a presentation for the superintendent.

- During Rachel's virtual tour of the VSF archive, Carlisle's display shows only what Rachel chooses to open. This makes it easy to show Ralph's judging forms, and the issues in comparing exhibits.
- To create her presentation, Rachel goes back through the winning exhibits, which are easy to find because of the blue/red/yellow ribbons attached to them.
- Opening each one, she looks over the [components](#) to choose the more visual elements, then opens them and makes copies of interesting displays.
- When she makes a copy, a message box reminds her that a copy notice was sent to the author.

tuation Feature	Possible Pros (+) or Cons (-) of the Feature	Scenarios
A main viewing window for each exhibit	⊕ directs viewers attention to the selected component ⊕ builds on the visual structure of the overall fair and of MOOsburg ⊖ but makes it impossible to view multiple components together	■ <u>Sally plans her exhibit on black holes.</u> ■ <u>Alicia and Delia go to the science fair.</u>
Labeling template structures with project subsections	⊕ reminds students of the science content expected in their exhibits ⊖ but may confuse visitors without science fair backgrounds ⊖ but may discourage innovative exhibit content	■ <u>Sally plans her exhibit on black holes.</u>
Embedded examples of expected components	⊕ gives concrete and direct insight into what can be included as an element ⊖ but students may worry if they do not have these component types	■ <u>Sally plans her exhibit on black holes.</u>
Nested exhibit components	⊕ simplifies browsing and comprehension of the top-level components ⊖ but viewers may not realize that these layers exist	■ <u>Mr. King coaches Sally on her project.</u>
Separate "windows" for the various exhibit components	⊕ aids perception and comprehension of the different activities ⊖ but may be confusing for exhibits with many components	■ <u>Sally plans her exhibit on black holes.</u> ■ <u>Rachel organizes a presentation for the superintendent.</u>
Using a room panorama for laying out exhibits	⊕ leverages visitors' familiarity with real world buildings and layouts ⊕ produces a convenient (spatial) overview of exhibits at	■ <u>Alicia and Delia go to the science fair.</u>

	the fair ⊖ but visitors may wonder if they must "walk" to reach exhibits	
Yellow and black flags over incomplete exhibits	⊕ builds on common experience with "under construction" sites ⊕ emphasizes that the fair activity is extended, dynamic, and ongoing ⊖ but these bright color cues make the display more complex	■ <u>Alicia and Delia go to the science fair.</u>
Listing all visitors to an exhibit	⊕ emphasizes the group- and community-oriented nature of this event ⊕ aids awareness of and access to co-present participants ⊖ but some visitors may object to having their "presence" advertised	■ <u>Sally plans her exhibit on black holes.</u> ■ <u>Alicia and Delia go to the science fair.</u>
Exhibit components shown as miniaturized windows	⊕ suggests that they may contain interactive content ⊖ but viewers may misinterpret them as independent applications	■ <u>Alicia and Delia go to the science fair.</u>
Customized VSF interfaces for judges	⊕ implies that judges are recognized and that their task is appreciated ⊕ simplifies access to judging-relevant information ⊖ but may reduce judges' connections to other activities in the fair	■ <u>Ralph judges the high school physics projects at the science fair.</u>
Hiding the calculations used to tally ratings	⊕ simplifies the form, directing attention to the main job of judging ⊖ but iudaes mav not realize that these formulas can be	■ <u>Ralph judges the high school physics projects at the science fair.</u>

edited